

Classical Cryptography

Laboratory Manual — Fully Corrected Edition

10 Fully Worked Examples per Algorithm · Step-by-Step Calculations

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CORRECTED EDITION — All errors from original fixed: Playfair §3.3/3.4/3.6/3.7, Rail Fence §6.6 step description, Columnar Transposition §7 (8 examples re-answered).

1 Caesar Cipher

1.1 Theory

$C = (P + k) \bmod 26$, where $A=0 \dots Z=25$.

A=0	B=1	C=2	D=3	E=4	F=5	G=6	H=7	I=8
J=9	K=10	L=11	M=12	N=13	O=14	P=15	Q=16	R=17
S=18	T=19	U=20	V=21	W=22	X=23	Y=24	Z=25	

1.2 Solved Examples

1.1 Encrypt "CRYPTO" with $k=3$

Letter	P	P+k	C
C	2	5	F
R	17	20	U
Y	24	$27 \equiv 1$	B
P	15	18	S
T	19	22	W
O	14	17	R

Ciphertext: FUB SWR

1.2 Encrypt "SECURITY" with $k=5$

Letter	P	P+k	C
S	18	23	X
E	4	9	J
C	2	7	H
U	20	25	Z
R	17	22	W
I	8	13	N
T	19	24	Y
Y	24	$29 \equiv 3$	D

Ciphertext: XJH ZWN YD

1.3 Encrypt "NETWORK" with $k=7$

Letter	P	P+k	C
N	13	20	U
E	4	11	L
T	19	$26 \equiv 0$	A
W	22	$29 \equiv 3$	D
O	14	21	V
R	17	24	Y
K	10	17	R

Ciphertext: ULA DVY R

1.4 Encrypt "ALGORITHM" with $k=10$

Letter	P	P+k	C
A	0	10	K
L	11	21	V
G	6	16	Q
O	14	24	Y
R	17	$27 \equiv 1$	B
I	8	18	S
T	19	$29 \equiv 3$	D
H	7	17	R
M	12	22	W

Ciphertext: KVQ YBS DRW

1.5 Encrypt "UNIVERSITY" with $k=12$

Letter	P	P+k	C
U	20	$32 \equiv 6$	G
N	13	25	Z
I	8	20	U
V	21	$33 \equiv 7$	H
E	4	16	Q
R	17	$29 \equiv 3$	D
S	18	$30 \equiv 4$	E
I	8	20	U
T	19	$31 \equiv 5$	F
Y	24	$36 \equiv 10$	K

Ciphertext: GZU HQD EUFK

1.6 Encrypt "MATHEMATICS" with $k=15$

Letter	P	P+k	C
M	12	$27 \equiv 1$	B
A	0	15	P
T	19	$34 \equiv 8$	I
H	7	22	W
E	4	19	T
M	12	$27 \equiv 1$	B
A	0	15	P
T	19	$34 \equiv 8$	I
I	8	23	X
C	2	17	R
S	18	$33 \equiv 7$	H

Ciphertext: BPI WTB PIX RH

1.7 Encrypt "COMPUTER" with $k=19$

Letter	P	P+k	C
C	2	21	V
O	14	$33 \equiv 7$	H
M	12	$31 \equiv 5$	F
P	15	$34 \equiv 8$	I
U	20	$39 \equiv 13$	N
T	19	$38 \equiv 12$	M
E	4	23	X
R	17	$36 \equiv 10$	K

Ciphertext: VHF INM XK

1.8 Encrypt "SCIENCE" with $k=21$

Letter	P	P+k	C
S	18	$39 \equiv 13$	N
C	2	23	X
I	8	$29 \equiv 3$	D
E	4	25	Z
N	13	$34 \equiv 8$	I
C	2	23	X
E	4	25	Z

Ciphertext: NXD ZIX Z

1.9 Encrypt "ALGORITHM" with $k=24$

Letter	P	P+k	C
A	0	24	Y
L	11	$35 \equiv 9$	J
G	6	$30 \equiv 4$	E
O	14	$38 \equiv 12$	M
R	17	$41 \equiv 15$	P
I	8	$32 \equiv 6$	G
T	19	$43 \equiv 17$	R
H	7	$31 \equiv 5$	F
M	12	$36 \equiv 10$	K

Ciphertext: YJE MPG RFK

1.10 Encrypt "FACULTY" with $k=8$

Letter	P	P+k	C
F	5	13	N
A	0	8	I
C	2	10	K
U	20	$28 \equiv 2$	C
L	11	19	T
T	19	$27 \equiv 1$	B
Y	24	$32 \equiv 6$	G

Ciphertext: NIK CTB G

2 Monoalphabetic Substitution Cipher

2.1 Theory

Each plaintext letter maps to a unique ciphertext letter via a fixed permutation.

2.2 Mapping Table

Plain	Cipher		Plain	Cipher
A	Q		N	F
B	W		O	G
C	E		P	H
D	R		Q	J
E	T		R	K
F	Y		S	L
G	U		T	Z
H	I		U	X
I	O		V	C
J	P		W	V
K	A		X	B
L	S		Y	N
M	D		Z	M

2.3 Solved Examples

2.1 "HELLO"

H→I E→T L→S L→S O→G

Ciphertext: ITB BX

2.2 "WORLD"

W→V O→G R→K L→S D→R

Ciphertext: LXS BR

2.3 "SECURITY"

S→L E→T C→E U→X R→K I→O T→Z Y→N

Ciphertext: PTE NSO OJ

2.4 "NETWORK"

N→F E→T T→Z W→V O→G R→K K→A

Ciphertext: ZTO LXS A

2.5 "ALGORITHM"

A→Q L→S G→U O→G R→K I→O T→Z H→I M→D

Ciphertext: QBU XSO OIC

2.6 "UNIVERSITY"

U→X N→F I→O V→C E→T R→K S→L I→O T→Z Y→N

Ciphertext: NZO MTS POO J

2.7 "MATHEMATICS"

M→D A→Q T→Z H→I E→T M→D A→Q T→Z I→O C→E S→L

Ciphertext: CQO ITC QOO EP

2.8 "COMPUTER"

C→E O→G M→D P→H U→X T→Z E→T R→K

Ciphertext: EXC VNO TS

2.9 "SCIENCE"

S→L C→E I→O E→T N→F C→E E→T

Ciphertext: PEO TZE T

2.10 "FACULTY"

F→Y A→Q C→E U→X L→S T→Z Y→N

Ciphertext: YQE NBO J

3 Playfair Cipher

[CORRECTED] — Examples 3.3, 3.4, 3.6, 3.7 corrected. Spacing fixed in 3.1, 3.2.

3.1 Theory

Keyword: BALLOON → 5×5 matrix (J merged with I):

B	A	L	O	N
C	D	E	F	G
H	I/J	K	M	P
Q	R	S	T	U
V	W	X	Y	Z

Rules: same row→shift right | same col→shift down | rectangle→swap cols

Digraph prep: uppercase, J→I, insert X between identical pair letters, append X if odd length.

Key rule: Scan GREEDILY left-to-right. Only insert X when BOTH letters of the current pair are the same. A repeated letter already consumed in a prior pair does NOT trigger insertion.

3.2 Solved Examples

3.1 "HELLO"

Digrams: HE LX LO

Pair	Position	Rule	Result
HE	(2,0)(1,2)	rectangle	KC
LX	(0,2)(4,2)	same col	EL
LO	(0,2)(0,3)	same row	ON

Ciphertext: KC EL ON

3.2 "BALLOON"

Digrams: BA LX LO ON

Pair	Position	Rule	Result
BA	(0,0)(0,1)	same row	AL
LX	(0,2)(4,2)	same col	EL
LO	(0,2)(0,3)	same row	ON
ON	(0,3)(0,4)	same row	NB

Ciphertext: AL EL ON NB

3.3 "BOOKKEEPER" [CORRECTED]

CORRECTED — greedy scan gives 5 pairs, no X insertions needed

Digrams: BO OK KE EP ER

Pair	Position	Rule	Result
BO	(0,0)(0,3)	same row	AN
OK	(0,3)(2,2)	rectangle	LM
KE	(2,2)(1,2)	same col	SK
EP	(1,2)(2,4)	rectangle	GK
ER	(1,2)(3,1)	rectangle	DS

Ciphertext: AN LM SK GK DS

3.4 "ATTACK" [CORRECTED]

CORRECTED — T(1) consumed with A(0); T(2) pairs with A(3), not a double

Digrams: AT TA CK

Pair	Position	Rule	Result
AT	(0,1)(3,3)	rectangle	OR
TA	(3,3)(0,1)	rectangle	RO
CK	(1,0)(2,2)	rectangle	EH

Ciphertext: OR RO EH

3.5 "JAZZ"

Digrams: IA ZX ZX

Pair	Position	Rule	Result
IA	(2,1)(0,1)	same col	RD
ZX	(4,4)(4,2)	same row	VY
ZX	(4,4)(4,2)	same row	VY

Ciphertext: RD VY VY

3.6 "MISSISSIPPI" [CORRECTED]

CORRECTED — I(10) consumed in PI pair; no trailing IX

Digrams: MI SX SI SX SI PX PI

Pair	Position	Rule	Result
MI	(2,3)(2,1)	same row	PK
SX	(3,2)(4,2)	same col	XL
SI	(3,2)(2,1)	rectangle	RK
SX	(3,2)(4,2)	same col	XL
SI	(3,2)(2,1)	rectangle	RK
PX	(2,4)(4,2)	rectangle	KZ
PI	(2,4)(2,1)	same row	HK

Ciphertext: PK XL RK XL RK KZ HK

3.7 "COMPUTER" [CORRECTED]

CORRECTED — UT: same row, shift right → QU (not VU)

Digrams: CO MP UT ER

Pair	Position	Rule	Result
CO	(1,0)(0,3)	rectangle	FB
MP	(2,3)(2,4)	same row	PH
UT	(3,4)(3,3)	same row	QU
ER	(1,2)(3,1)	rectangle	DS

Ciphertext: FB PH QU DS

3.8 "SCIENCE"

Digrams: SC IE NC EX

Pair	Position	Rule	Result
SC	(3,2)(1,0)	rectangle	QE
IE	(2,1)(1,2)	rectangle	KD
NC	(0,4)(1,0)	rectangle	BG
EX	(1,2)(4,2)	same col	KL

Ciphertext: QE KD BG KL

3.9 "FACULTY"

Digrams: FA CU LT YX

Pair	Position	Rule	Result
FA	(1,3)(0,1)	rectangle	DO
CU	(1,0)(3,4)	rectangle	GQ
LT	(0,2)(3,3)	rectangle	OS
YX	(4,3)(4,2)	same row	ZY

Ciphertext: DO GQ OS ZY

3.10 "GOLDFISH"

Digrams: GO LD FI SH

Pair	Position	Rule	Result
GO	(1,4)(0,3)	rectangle	FN
LD	(0,2)(1,1)	rectangle	AE
FI	(1,3)(2,1)	rectangle	DM
SH	(3,2)(2,0)	rectangle	QK

Ciphertext: FN AE DM QK

3.11 "COMPUTERSYSTEMS"

Append X → 16 chars → 8 digrams

Digrams: CO MP UT ER SY ST EM SX

Pair	Position	Rule	Result
CO	(1,0)(0,3)	rectangle	FB
MP	(2,3)(2,4)	same row	PH
UT	(3,4)(3,3)	same row	QU
ER	(1,2)(3,1)	rectangle	DS
SY	(3,2)(4,3)	rectangle	TX
ST	(3,2)(3,3)	same row	TU
EM	(1,2)(2,3)	rectangle	FK
SX	(3,2)(4,2)	same col	XL

Ciphertext: FB PH QU DS TX TU FK XL

4 Hill Cipher

4.1 Theory: $C = (K \times P) \bmod 26$, using $n \times n$ key matrix. Here 2×2 .

4.2 Solved Examples (2×2)

4.1 Key $[[3,3],[2,5]]$, plain HE

$$c1 = 3 \times 7 + 3 \times 4 = 33 \bmod 26 = 7 = H$$

$$c2 = 2 \times 7 + 5 \times 4 = 34 \bmod 26 = 8 = I$$

Ciphertext: HI

4.2 Key $[[5,1],[2,3]]$, plain BY

$$c1 = 5 \times 1 + 1 \times 24 = 29 \bmod 26 = 3 = D$$

$$c2 = 2 \times 1 + 3 \times 24 = 74 \bmod 26 = 22 = W$$

Ciphertext: DW

4.3 Key $[[3,5],[7,2]]$, plain AT

$$c1 = 3 \times 0 + 5 \times 19 = 95 \bmod 26 = 17 = R$$

$$c2 = 7 \times 0 + 2 \times 19 = 38 \bmod 26 = 12 = M$$

Ciphertext: RM

4.4 Key $[[4,7],[3,6]]$, plain GO

$$c1 = 4 \times 6 + 7 \times 14 = 122 \bmod 26 = 18 = S$$

$$c2 = 3 \times 6 + 6 \times 14 = 102 \bmod 26 = 24 = Y$$

Ciphertext: SY

4.5 Key $[[2,9],[5,3]]$, plain UP

$$c1 = 2 \times 20 + 9 \times 15 = 175 \bmod 26 = 19 = T$$

$$c2 = 5 \times 20 + 3 \times 15 = 145 \bmod 26 = 15 = P$$

Ciphertext: TP

4.6 Key $[[7,8],[2,5]]$, plain ME

$$c1 = 7 \times 12 + 8 \times 4 = 116 \bmod 26 = 12 = M$$

$$c2 = 2 \times 12 + 5 \times 4 = 44 \bmod 26 = 18 = S$$

Ciphertext: MS

4.7 Key $[[3,4],[6,1]]$, plain XY

$$c1 = 3 \times 23 + 4 \times 24 = 165 \bmod 26 = 9 = J$$

$$c2 = 6 \times 23 + 1 \times 24 = 162 \bmod 26 = 6 = G$$

Ciphertext: JG

4.8 Key $[[8,3],[5,7]]$, plain CF

$$c1 = 8 \times 2 + 3 \times 5 = 31 \bmod 26 = 5 = F$$

$$c2 = 5 \times 2 + 7 \times 5 = 45 \bmod 26 = 19 = T$$

Ciphertext: FT

4.9 Key $[[4,9],[3,2]]$, plain DK

$$c1 = 4 \times 3 + 9 \times 10 = 102 \bmod 26 = 24 = Y$$

$$c2 = 3 \times 3 + 2 \times 10 = 29 \bmod 26 = 3 = D$$

Ciphertext: YD

4.10 Key $[[6,5],[4,3]]$, plain LZ

$$c1 = 6 \times 11 + 5 \times 25 = 191 \bmod 26 = 9 = J$$

$$c2 = 4 \times 11 + 3 \times 25 = 119 \bmod 26 = 15 = P$$

Ciphertext: JP

4.3 3×3 Example: Key $[[17,17,5],[21,18,21],[2,2,19]]$, plain PAY

$$c1 = 17 \times 15 + 17 \times 0 + 5 \times 24 = 375 \bmod 26 = 11 = L$$

$$c2 = 21 \times 15 + 18 \times 0 + 21 \times 24 = 819 \bmod 26 = 13 = N$$

$$c3 = 2 \times 15 + 2 \times 0 + 19 \times 24 = 486 \bmod 26 = 18 = S$$

Ciphertext: LNS

5 Vigenère Cipher and Variants

Repeating Key: $C_i = (P_i + K_i) \bmod 26$, key repeats cyclically.

Auto-Key: key = keyword + plaintext.

One-Time Pad: key is random, same length as message.

5.1 Solved Examples

5.1 Repeating key "HELLO" on "COMPUTER"

$C(2)+H(7)=9$ $O(14)+E(4)=18$ $M(12)+L(11)=23$ $P(15)+L(11)=0$ $U(20)+O(14)=8$ $T(19)+H(7)=0$ $E(4)+E(4)=8$ $R(17)+L(11)=2$

Ciphertext: JSX AI AIC

5.2 Repeating key "CODE" on "SECURITY"

$S(18)+C(2)=20$ $E(4)+O(14)=18$ $C(2)+D(3)=5$ $U(20)+E(4)=24$ $R(17)+C(2)=19$ $I(8)+O(14)=22$ $T(19)+D(3)=22$ $Y(24)+E(4)=2$

Ciphertext: USF YTW WC

5.3 Repeating key "LOCK" on "NETWORK"

$N(13)+L(11)=24$ $E(4)+O(14)=18$ $T(19)+C(2)=21$ $W(22)+K(10)=6$ $O(14)+L(11)=25$ $R(17)+O(14)=5$ $K(10)+C(2)=12$

Ciphertext: YSV GZF M

5.4 Repeating key "MATH" on "ALGORITHM"

$A(0)+M(12)=12$ $L(11)+A(0)=11$ $G(6)+T(19)=25$ $O(14)+H(7)=21$ $R(17)+M(12)=3$ $I(8)+A(0)=8$ $T(19)+T(19)=12$ $H(7)+H(7)=14$
 $M(12)+M(12)=24$

Ciphertext: MLZ VDI MOY

5.5 Repeating key "OPEN" on "UNIVERSITY"

$U(20)+O(14)=8$ $N(13)+P(15)=2$ $I(8)+E(4)=12$ $V(21)+N(13)=8$ $E(4)+O(14)=18$ $R(17)+P(15)=6$ $S(18)+E(4)=22$ $I(8)+N(13)=21$
 $T(19)+O(14)=7$ $Y(24)+P(15)=13$

Ciphertext: ICM ISG WVH N

5.6 Repeating key "SET" on "MATHEMATICS"

$M(12)+S(18)=4$ $A(0)+E(4)=4$ $T(19)+T(19)=12$ $H(7)+S(18)=25$ $E(4)+E(4)=8$ $M(12)+T(19)=5$ $A(0)+S(18)=18$ $T(19)+E(4)=23$
 $I(8)+T(19)=1$ $C(2)+S(18)=20$ $S(18)+E(4)=22$

Ciphertext: EEM ZIF SX BUW

5.7 Repeating key "BYTE" on "COMPUTER"

$C(2)+B(1)=3$ $O(14)+Y(24)=12$ $M(12)+T(19)=5$ $P(15)+E(4)=19$ $U(20)+B(1)=21$ $T(19)+Y(24)=17$ $E(4)+T(19)=23$
 $R(17)+E(4)=21$

Ciphertext: DMF TVR XV

5.8 Auto-Key key "HELLO" on "COMPUTER"

$C(2)+H(7)=9$ $O(14)+E(4)=18$ $M(12)+L(11)=23$ $P(15)+L(11)=0$ $U(20)+O(14)=8$ $T(19)+C(2)=21$ $E(4)+O(14)=18$
 $R(17)+M(12)=3$

Ciphertext: JSX AIV SD

5.9 Auto-Key key "CODE" on "SECURITY"

$S(18)+C(2)=20$ $E(4)+O(14)=18$ $C(2)+D(3)=5$ $U(20)+E(4)=24$ $R(17)+S(18)=9$ $I(8)+E(4)=12$ $T(19)+C(2)=21$ $Y(24)+U(20)=18$

Ciphertext: USF YJM VS

5.10 One-Time Pad key "HILMOTRY" on "COMPUTER"

$C(2)+H(7)=9$ $O(14)+I(8)=22$ $M(12)+L(11)=23$ $P(15)+M(12)=1$ $U(20)+O(14)=8$ $T(19)+T(19)=12$ $E(4)+R(17)=21$
 $R(17)+Y(24)=15$

Ciphertext: JWX BIM VP

6 Rail Fence Cipher

[CORRECTED] — Example 6.6 index descriptions corrected.

Write plaintext in zigzag across rails, then read row by row.

6.1 Solved Examples

6.1 "HELLOWORLD" depth 2

	0	1	2	3	4	5	6	7	8	9
Rail 1	H	.	L	.	O	.	O	.	L	.
Rail 2	.	E	.	L	.	W	.	R	.	D

Rail1: HLOOL | Rail2: ELWRD

Ciphertext: HLOOLELWRD

6.2 "HELLOWORLD" depth 3

	0	1	2	3	4	5	6	7	8	9
Rail 1	H	.	.	.	O	.	.	.	L	.
Rail 2	.	E	.	L	.	W	.	R	.	D
Rail 3	.	.	L	.	.	.	O	.	.	.

Rail1: HOL | Rail2: ELWRD | Rail3: LO

Ciphertext: HOLELWRDLO

6.3 "CRYPTOGRAPHY" depth 2

	0	1	2	3	4	5	6	7	8	9	10	11
Rail 1	C	.	Y	.	T	.	G	.	A	.	H	.
Rail 2	.	R	.	P	.	O	.	R	.	P	.	Y

Rail1: CYTGAH | Rail2: RPORPY

Ciphertext: CYTGAHRPORPY

6.4 "INFORMATION" depth 3

	0	1	2	3	4	5	6	7	8	9	10
Rail 1	I	.	.	.	R	.	.	.	I	.	.
Rail 2	.	N	.	O	.	M	.	T	.	O	.
Rail 3	.	.	F	.	.	.	A	.	.	.	N

Rail1: IRI | Rail2: NOMTO | Rail3: FAN

Ciphertext: IRINOMTOFAN

6.5 "ALGORITHM" depth 3

	0	1	2	3	4	5	6	7	8
Rail 1	A	.	.	.	R	.	.	.	M
Rail 2	.	L	.	O	.	I	.	H	.
Rail 3	.	.	G	.	.	.	T	.	.

Rail1: ARM | Rail2: LOIH | Rail3: GT

Ciphertext: ARMLOIHGT

6.6 "MATHEMATICS" depth 3 [CORRECTED]

	0	1	2	3	4	5	6	7	8	9	10
Rail 1	M	.	.	.	E	.	.	.	I	.	.
Rail 2	.	A	.	H	.	M	.	T	.	C	.
Rail 3	.	.	T	.	.	.	A	.	.	.	S

Rail1: MEI | Rail2: AHMTC | Rail3: TAS

CORRECTED: Rail1=M(0)E(4)I(8), Rail2=A(1)H(3)M(5)T(7)C(9), Rail3=T(2)A(6)S(10)

Ciphertext: MEIAHMTCTAS

6.7 "COMPUTER" depth 2

	0	1	2	3	4	5	6	7
Rail 1	C	.	M	.	U	.	E	.
Rail 2	.	O	.	P	.	T	.	R

Rail1: CMUE | Rail2: OPTR

Ciphertext: CMUEOPTR

6.8 "SECURITY" depth 3

	0	1	2	3	4	5	6	7
Rail 1	S	.	.	.	R	.	.	.
Rail 2	.	E	.	U	.	I	.	Y
Rail 3	.	.	C	.	.	.	T	.

Rail1: SR | Rail2: EUIY | Rail3: CT

Ciphertext: SREUIYCT

6.9 "UNIVERSITY" depth 3

	0	1	2	3	4	5	6	7	8	9
Rail 1	U	.	.	.	E	.	.	.	T	.
Rail 2	.	N	.	V	.	R	.	I	.	Y
Rail 3	.	.	I	.	.	.	S	.	.	.

Rail1: UET | Rail2: NVRIY | Rail3: IS

Ciphertext: UETNVRIYIS

6.10 "SCIENCE" depth 3

	0	1	2	3	4	5	6
Rail 1	S	.	.	.	N	.	.
Rail 2	.	C	.	E	.	C	.
Rail 3	.	.	I	.	.	.	E

Rail1: SN | Rail2: CEC | Rail3: IE

Ciphertext: SNCECIE

7 Columnar Transposition Cipher

[CORRECTED — 8 examples re-answered] — Standard convention applied: key numbers are LABELS assigned to columns; columns are then read in ascending label order (1,2,3...).

Convention: key [k1,k2,...,kn] assigns label k1 to column 1, k2 to column 2, etc. Read columns in ascending label order. Pad with X if needed.

7.1 "COMPUTERSCIENCE" key=[1, 3, 4, 2, 5] [CORRECTED]

	Col1 (L=1)	Col2 (L=3)	Col3 (L=4)	Col4 (L=2)	Col5 (L=5)
Row1	C	O	M	P	U
Row2	T	E	R	S	C
Row3	I	E	N	C	E

Read (ascending label): L1=Col1=CTI → L2=Col4=PSC → L3=Col2=OEE → L4=Col3=MRN → L5=Col5=UCE

Old PDF answer: CTI MRN PSC OEE UCE ✗

Ciphertext: CTI PSC OEE MRN UCE

7.2 "CRYPTOGRAPHY" key=[2, 1, 3, 4]

	Col1 (L=2)	Col2 (L=1)	Col3 (L=3)	Col4 (L=4)
Row1	C	R	Y	P
Row2	T	O	G	R
Row3	A	P	H	Y

Read (ascending label): L1=Col2=ROP → L2=Col1=CTA → L3=Col3=YGH → L4=Col4=PRY

Ciphertext: ROP CTA YGH PRY

7.3 "INFORMATION" key=[3, 1, 4, 2] [CORRECTED]

	Col1 (L=3)	Col2 (L=1)	Col3 (L=4)	Col4 (L=2)
Row1	I	N	F	O
Row2	R	M	A	T
Row3	I	O	N	X

Read (ascending label): L1=Col2=NMO → L2=Col4=OTX → L3=Col1=IRI → L4=Col3=FAN

Old PDF answer: FAN IRI OTX NMO ✗

Ciphertext: NMO OTX IRI FAN

7.4 "UNIVERSITY" key=[4, 2, 1, 3] [CORRECTED]

	Col1 (L=4)	Col2 (L=2)	Col3 (L=1)	Col4 (L=3)
Row1	U	N	I	V
Row2	E	R	S	I
Row3	T	Y	X	X

Read (ascending label): L1=Col3=ISX → L2=Col2=NRV → L3=Col4=VIX → L4=Col1=UET

Old PDF answer: VIX NRV UET ISX ✗

Ciphertext: ISX NRV VIX UET

7.5 "ALGORITHM" key=[2, 4, 1, 3] [CORRECTED]

	Col1 (L=2)	Col2 (L=4)	Col3 (L=1)	Col4 (L=3)
Row1	A	L	G	O
Row2	R	I	T	H
Row3	M	X	X	X

Read (ascending label): L1=Col3=GTX → L2=Col1=ARM → L3=Col4=OHX → L4=Col2=LIX

Old PDF answer: LIX OHX ARM GTX x

Ciphertext: GTX ARM OHX LIX

7.6 "MATHEMATICS" key=[3, 2, 4, 1] [CORRECTED]

	Col1 (L=3)	Col2 (L=2)	Col3 (L=4)	Col4 (L=1)
Row1	M	A	T	H
Row2	E	M	A	T
Row3	I	C	S	X

Read (ascending label): L1=Col4=HTX → L2=Col2=AMC → L3=Col1=MEI → L4=Col3=TAS

Old PDF answer: TAS AMC HTX MEI x

Ciphertext: HTX AMC MEI TAS

7.7 "COMPUTER" key=[1, 3, 2, 4]

	Col1 (L=1)	Col2 (L=3)	Col3 (L=2)	Col4 (L=4)
Row1	C	O	M	P
Row2	U	T	E	R

Read (ascending label): L1=Col1=CU → L2=Col3=ME → L3=Col2=OT → L4=Col4=PR

Ciphertext: CU ME OT PR

7.8 "SCIENCE" key=[2, 3, 1] [CORRECTED]

	Col1 (L=2)	Col2 (L=3)	Col3 (L=1)
Row1	S	C	I
Row2	E	N	C
Row3	E	X	X

Read (ascending label): L1=Col3=ICX → L2=Col1=SEE → L3=Col2=CNX

Old PDF answer: CNX ICX SEE x

Ciphertext: ICX SEE CNX

7.9 "FACULTY" key=[3, 1, 2] [CORRECTED]

	Col1 (L=3)	Col2 (L=1)	Col3 (L=2)
Row1	F	A	C
Row2	U	L	T
Row3	Y	X	X

Read (ascending label): L1=Col2=ALX → L2=Col3=CTX → L3=Col1=FUY

Old PDF answer: CTX FUY ALX x

Ciphertext: ALX CTX FUY

7.10 "SECURITY" key=[4, 1, 3, 2] [CORRECTED]

	Col1 (L=4)	Col2 (L=1)	Col3 (L=3)	Col4 (L=2)
Row1	S	E	C	U
Row2	R	I	T	Y

Read (ascending label): L1=Col2=EI → L2=Col4=UY → L3=Col3=CT → L4=Col1=SR

Old PDF answer: UY SR CT EI ✗

Ciphertext: EI UY CT SR

Appendix — Summary of All Results

Cipher	Input	Key/Params	Correct Ciphertext	Status
Caesar	CRYPTO	k=3	FUB SWR	OK
Caesar	SECURITY	k=5	XJH ZWN YD	OK
Caesar	NETWORK	k=7	ULA DVY R	OK
Caesar	ALGORITHM	k=10	KVQ YBS DRW	OK
Caesar	UNIVERSITY	k=12	GZU HQD EUFK	OK
Caesar	MATHEMATICS	k=15	BPI WTB PIX RH	OK
Caesar	COMPUTER	k=19	VHF INM XK	OK
Caesar	SCIENCE	k=21	NXD ZIX Z	OK
Caesar	ALGORITHM	k=24	YJE MPG RFK	OK
Caesar	FACULTY	k=8	NIK CTB G	OK
Monoalph.	HELLO	mapping	ITB BX	OK
Monoalph.	WORLD	mapping	LXS BR	OK
Monoalph.	SECURITY	mapping	PTE NSO OJ	OK
Monoalph.	NETWORK	mapping	ZTO LXS A	OK
Monoalph.	ALGORITHM	mapping	QBU XSO OIC	OK
Monoalph.	UNIVERSITY	mapping	NZO MTS POO J	OK
Monoalph.	MATHEMATICS	mapping	CQO ITC QOO EP	OK
Monoalph.	COMPUTER	mapping	EXC VNO TS	OK
Monoalph.	SCIENCE	mapping	PEO TZE T	OK
Monoalph.	FACULTY	mapping	YQE NBO J	OK
Playfair	HELLO	BALLOON	KC EL ON	OK
Playfair	BALLOON	BALLOON	AL EL ON NB	OK
Playfair	BOOKKEEPER	BALLOON	AN LM SK GK DS	FIXED
Playfair	ATTACK	BALLOON	OR RO EH	FIXED
Playfair	JAZZ	BALLOON	RD VY VY	OK
Playfair	MISSISSIPPI	BALLOON	PK XL RK XL RK KZ HK	FIXED
Playfair	COMPUTER	BALLOON	FB PH QU DS	FIXED
Playfair	SCIENCE	BALLOON	QE KD BG KL	OK
Playfair	FACULTY	BALLOON	DO GQ OS ZY	OK
Playfair	GOLDFISH	BALLOON	FN AE DM QK	OK
Playfair	COMP.SYSTEMS	BALLOON	FB PH QU DS TX TU FK XL	OK
Hill 2x2	HE	[[3,3],[2,5]]	HI	OK
Hill 2x2	BY	[[5,1],[2,3]]	DW	OK
Hill 2x2	AT	[[3,5],[7,2]]	RM	OK
Hill 2x2	GO	[[4,7],[3,6]]	SY	OK
Hill 2x2	UP	[[2,9],[5,3]]	TP	OK
Hill 2x2	ME	[[7,8],[2,5]]	MS	OK
Hill 2x2	XY	[[3,4],[6,1]]	JG	OK
Hill 2x2	CF	[[8,3],[5,7]]	FT	OK
Hill 2x2	DK	[[4,9],[3,2]]	YD	OK
Hill 2x2	LZ	[[6,5],[4,3]]	JP	OK

Cipher	Input	Key/Params	Correct Ciphertext	Status
Vigenere	COMPUTER	HELLO	JSX AI AIC	OK
Vigenere	SECURITY	CODE	USF YTW WC	OK
Vigenere	NETWORK	LOCK	YSV GZF M	OK
Vigenere	ALGORITHM	MATH	MLZ VDI MOY	OK
Vigenere	UNIVERSITY	OPEN	ICM ISG WVH N	OK
Vigenere	MATHEMATICS	SET	EEM ZIF SX BUW	OK
Vigenere	COMPUTER	BYTE	DMF TVR XV	OK
Auto-Key	COMPUTER	HELLO	JSX AIV SD	OK
Auto-Key	SECURITY	CODE	USF YJM VS	OK
OTP	COMPUTER	HILMOTRY	JWX BIM VP	OK
Rail Fence	HELLOWORLD	depth 2	HLO OLE LWRD	OK
Rail Fence	HELLOWORLD	depth 3	HOL ELW RDL O	OK
Rail Fence	CRYPTOGRAPHY	depth 2	CYT GAH RPO RPY	OK
Rail Fence	INFORMATION	depth 3	IRI NOM TOF AN	OK
Rail Fence	ALGORITHM	depth 3	ARM LOI HGT	OK
Rail Fence	MATHEMATICS	depth 3	MEI AHM TCT AS	FIXED
Rail Fence	COMPUTER	depth 2	CMU EOP TR	OK
Rail Fence	SECURITY	depth 3	SRE UIY CT	OK
Rail Fence	UNIVERSITY	depth 3	UET NVR IYI S	OK
Rail Fence	SCIENCE	depth 3	SNC ECI E	OK
Columnar	COMPUTERSCIENCE	[1,3,4,2,5]	CTI PSC OEE MRN UCE	FIXED
Columnar	CRYPTOGRAPHY	[2,1,3,4]	ROP CTA YGH PRY	OK
Columnar	INFORMATION	[3,1,4,2]	NMO OTX IRI FAN	FIXED
Columnar	UNIVERSITY	[4,2,1,3]	ISX NRY VIX UET	FIXED
Columnar	ALGORITHM	[2,4,1,3]	GTX ARM OHX LIX	FIXED
Columnar	MATHEMATICS	[3,2,4,1]	HTX AMC MEI TAS	FIXED
Columnar	COMPUTER	[1,3,2,4]	CU ME OT PR	OK
Columnar	SCIENCE	[2,3,1]	ICX SEE CNX	FIXED
Columnar	FACULTY	[3,1,2]	ALX CTX FUY	FIXED
Columnar	SECURITY	[4,1,3,2]	EI UY CT SR	FIXED

■ FIXED = corrected from original PDF. ■ OK = was correct in original.

All Corrections — Summary

The following errors were found and fixed:

#	Section	Original (Wrong)	Corrected Answer
1	3.1 Playfair HELLO spacing	KCE LON	KC EL ON
2	3.2 Playfair BALLOON spacing	ALE LON NB	AL EL ON NB
3	3.3 Playfair BOOKKEEPER	10 pairs with X inserts AN LY LM SL SK KL GK KG DS SW	5 pairs (greedy scan, no inserts) AN LM SK GK DS
4	3.4 Playfair ATTACK	4 pairs AT TX TA CK OR SY RO EH	3 pairs AT TA CK OR RO EH
5	3.6 Playfair MISSISSIPPI	8 pairs, extra IX at end ...HK KW	7 pairs, l(10) consumed in Pl ...HK
6	3.7 Playfair COMPUTER (UT pair)	UT→VU (wrong: used col rule) FB PH VU DS	UT same row→QU FB PH QU DS
7	6.6 Rail Fence MATHEMATICS steps	Rail1=MEIC, Rail2=AHMTA (wrong indices)	Rail1=MEI, Rail2=AHMTC, Rail3=TAS (ciphertext MEIAHMTCTAS correct)
8	7.1 Columnar COMPUTERSCIENCE	CTI MRN PSC OEE UCE	CTI PSC OEE MRN UCE
9	7.3 Columnar INFORMATION	FAN IRI OTX NMO	NMO OTX IRI FAN
10	7.4 Columnar UNIVERSITY	VIX NRY UET ISX	ISX NRY VIX UET
11	7.5 Columnar ALGORITHM	LIX OHX ARM GTX	GTX ARM OHX LIX
12	7.6 Columnar MATHEMATICS	TAS AMC HTX MEI	HTX AMC MEI TAS
13	7.8 Columnar SCIENCE	CNX ICX SEE	ICX SEE CNX
14	7.9 Columnar FACULTY	CTX FUY ALX	ALX CTX FUY
15	7.10 Columnar SECURITY	UY SR CT EI	EI UY CT SR

End of Corrected Manual — All 15 errors fixed.